Pravar Mishra

Backend Engineer · Applied Vision ML · Model Fine-Tuning

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PROFILE

Software/ML engineer (B.Tech CSE, GATE qualified) working on **computer vision for satellite SAR imagery**. Works with operational **ICEYE** products for defence users, previously shipped satellite-imagery ML services at **Dhruva Space** and built a backend system prototype at **NRSC, ISRO**. Trained a **multi-dataset YOLO11m ship detector** and built SARnotate, an annotation and detection platform for ICEYE scenes.

TECHNICAL SKILLS

ML & Vision	PyTorch, fine-tuning for specialised domains, object detection (YOLO11, CountGD / GroundingDINO-based), CNN classification, exemplar-conditioned detection, multi-dataset training, error analysis
Remote Sensing	SAR (ICEYE GRD, Sentinel-1), speckle-aware signal processing, autocorrelation, SAR georeferencing (affine, GCP, RPC)
Geospatial	GeoTIFF / Cloud-Optimized GeoTIFF, rasterio, STAC / pgstac, TiTiler, GeoJSON
Pipelines & Serving	Python, Java, FastAPI, tiled batched GPU inference, model serving, PostgreSQL, Spring Security

EXPERIENCE

SAR Image Consultant · ICEYE, Espoo, Finland (Remote) Jul 2026 – Present

- Work with operational products from ICEYE's satellite constellation, supporting **target interpretation for defence and intelligence use cases**; trained on-site at ICEYE HQ in Espoo.
- Built **SARnotate**, an annotation and detection platform that produces **provenance-tagged training data**: every box records whether a human, model, copy or automated tool created it, keeping human-verified labels separable to **avoid circular supervision**.
- Designed **lattice placement** for vehicle depots that works around SAR speckle. It measures object pitch by autocorrelation (speckle defeats template matching), refines to sub-pixel, and places hundreds of boxes from one exemplar.
- Helped the team in interpreting targets of interest.

Software Engineer Intern · Dhruva Space, Hyderabad Jan 2025 – Apr 2025

- Built an **ML object-detection and counting service (CountGD)** for satellite imagery, used for automated feature extraction.
- Delivered a **map tile service (TiTiler)** and a **STAC catalogue API on PostgreSQL** serving large imagery archives; implemented TLE handling for collection planning.

Software Intern · NRSC, ISRO, Hyderabad Sep 2023 – Nov 2023

- Designed a prototype **authentication architecture (Spring Security 6)** adopted for NRSC applications, and built inter-service communication over HTTP.

RESEARCH & ML PROJECTS

SAR ship detector · YOLO11m · PyTorch · Independent research 2026

- Trained on a **merge of four SAR datasets (~14,800 images: HRSID, SSDD, LS-SSDD, Roboflow)**, reconciling COCO and VOC labels across a wide range of resolutions and scene contexts. Reached **0.906 mAP50** on held-out validation.
- **Transfers to ICEYE spotlight products absent from training**, the reason for choosing multi-resolution training over a single-dataset model. Runs tiled, batched GPU inference at a recall-oriented confidence floor.
- Found a preprocessing clip calibrated on ICEYE that truncates Sentinel-1's dynamic range; the fix is normalising before the 8-bit cast for **multi-sensor robustness**.

Vehicle Recognition CNN on MSTAR · Python, PyTorch · Independent research 2025 – 2026

- Built an end-to-end PyTorch training and evaluation pipeline for 10-class SAR vehicle recognition, reaching **96.5% accuracy** on a held-out test set with a **real distribution shift** from the training data.
- Diagnosed a dominant false-positive class from the confusion matrix and fixed it by capacity scaling (**errors 237 → 84**). Characterised confident-but-wrong predictions to set realistic performance baselines.

EDUCATION

B.Tech, Computer Science & Engineering · Vellore Institute of Technology, Andhra Pradesh · CGPA 8.75 / 102021 – 2025

GATE (Computer Science), Qualified · All India Rank 6366 · Score 520 · NCC 'A' Certificate2026

Languages · English (fluent), Hindi (native), Telugu